

1. Due to finite nuclear mass effect, the spectral lines of the deuterium ${}_1\text{H}^2$ is shifted compared to hydrogen. Estimate the shift for $\text{H}\alpha$ line.
2. *Spectral shift due to finite nuclear size*: Problem 5.9 in B&J
3. *Classical forbidden region and Quantum mechanics*: Problem 3.2 in B&J.
4. *Virial theorem*: Problem 3.1 in B&J.
5. Find the expression for the energy difference between the two extreme components of a fine structure multiplet.
6. *Second-order correction to the energy in perturbation theory*: Problem 5.6 in B&J (ignore Stark part).